

Notes: Nini, Smith, and Sufi (JFE, 2009)

Creditor control rights and firm investment policy

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1 Big picture

- **Question.** Do creditors directly control firm investment policy through private credit agreements, and do those controls affect investment and performance?
- **Core setting.** The paper studies 3,720 private credit agreements between banks and publicly traded U.S. firms over 1996–2005, collected from SEC EDGAR filings.
- **Main empirical fact.** Capital expenditure restrictions are common in private credit agreements: 32% of the agreements contain an explicit covenant restricting borrower capital expenditures.
- **Contrast with public bonds.** Earlier covenant studies of public bond indentures found very few direct restrictions on investment. The key difference is that private bank loans contain more detailed, tightly set, and renegotiable covenants.
- **Why restrictions exist.** The restrictions are motivated by creditor-shareholder conflicts. When credit quality deteriorates, shareholders have stronger incentives to take risky or value-reducing investments that transfer value from creditors.
- **Credit-quality result.** Restrictions become much more likely when borrower credit quality worsens. In particular, moving from investment grade to speculative grade sharply increases the probability of a capital expenditure restriction.
- **Contingency result.** The restrictions are not always written as explicit state-contingent clauses, but renegotiation after financial covenant violations makes them effectively contingent on borrower performance.
- **Investment result.** Firms receiving a new capital expenditure restriction reduce investment more than comparable firms without a restriction, even after controls for cash flow and investment opportunities.
- **Bindingness evidence.** Actual capital expenditures bunch just below the contractual limit, and bunching becomes much stronger after the restriction is introduced.
- **Performance result.** Firms that receive a new restriction experience subsequent increases in market-to-book and operating performance.
- **Economic takeaway.** Creditor control rights in private debt contracts are an important governance mechanism. They restrict managerial or shareholder investment choices, and the evidence suggests that these restrictions can improve firm value rather than merely ration investment.

2 Data and measurement

2.1 Private credit agreements

- The sample starts with loan deals from Reuters Loan Pricing Corporation’s DealScan database.
- The authors match the loans to firm financial characteristics from Compustat.
- They then search SEC EDGAR filings for the actual credit agreements, because the covenants and detailed restrictions are written in the contracts themselves.
- The final matched sample contains 3,720 private credit agreements to 1,931 borrowers over 1996–2005.
- The agreements are bank loan contracts, including sole-lender and syndicated facilities.

Interpretation: Private credit agreements are the right object because public bonds usually do not contain detailed direct investment restrictions, while private bank loans often do.

2.2 Capital expenditure restriction

The main treatment variable is an indicator:

$$\text{Restriction}_{it} = 1$$

if the credit agreement for firm i at time t contains a covenant that directly restricts borrower capital expenditures.

- A restriction can limit total capital expenditures over a fiscal year or over a specified period.
- Some contracts specify an exact dollar cap; the authors collect the limit amount for a subset of agreements.
- In the full sample, 32% of credit agreements contain a capital expenditure restriction.
- Among firms, 42% have a restriction at some point during the sample period.

Interpretation: The restriction is a direct creditor control right over the firm's investment policy, not merely an indirect financial covenant.

2.3 Restricted amount and bindingness

For 486 agreements, the authors collect the actual restricted dollar amount. For these observations, define

$$Gap_{it} = \frac{\text{CapEx}_{it} - \text{Limit}_{it}}{\text{Assets}_{i,t-1}}.$$

- $Gap_{it} < 0$ means actual capital expenditures are below the contractual limit.
- $Gap_{it} > 0$ means actual capital expenditures exceed the limit.
- The average restricted amount is 8.5% of lagged assets.
- Actual capital expenditures for the same subset average 6.4% of lagged assets.

Interpretation: If the restriction binds, expenditures should cluster just below zero in the distribution of Gap_{it} . This is exactly what the paper documents.

2.4 Credit-quality measures

The paper uses several measures of borrower credit quality:

- debt-to-cash-flow quartiles;
- negative cash flow indicator;
- Standard & Poor's issuer credit rating;
- financial covenant violation in the four quarters before the credit agreement.

Interpretation: These measures capture the idea that creditor-shareholder conflicts become stronger as debt becomes riskier.

2.5 Other contract terms

The paper compares capital expenditure restrictions with other loan contract terms:

- interest spread over LIBOR;
- collateral requirement / secured loan indicator;

- dividend restriction;
- coverage covenants;
- debt-to-cash-flow covenant;
- net worth covenant;
- debt-to-balance-sheet covenant;
- liquidity covenant;
- cash-flow-only covenant.

Interpretation: This comparison shows whether investment restrictions are just one minor covenant or whether they are as sensitive to credit quality as core loan terms such as spreads and collateral.

2.6 Investment and performance outcomes

The main investment outcome is annual capital expenditures scaled by lagged assets:

$$\frac{\text{CapEx}_{it}}{\text{Assets}_{i,t-1}}.$$

The main performance outcomes are:

$$MTB_{it} = \frac{\text{Market value of assets}}{\text{Book assets}}, \quad ROA_{it} = \frac{EBITDA}{\text{Lagged assets}}.$$

Interpretation: The paper asks not only whether restrictions reduce investment, but also whether the post-restriction firm looks better or worse.

3 Models and empirical specifications

3.1 Credit-quality determinants of restrictions

The paper begins with a general model for the probability of a capital expenditure restriction:

$$\Pr(\text{Restriction}_{it} = 1 \mid X_{it}\beta, c_i) = G(X_{it}\beta, c_i), \quad (1)$$

where X_{it} contains borrower credit-quality measures, loan controls, year indicators, purpose indicators, and loan-type indicators.

Interpretation: A positive coefficient on bad credit quality means creditors are more likely to restrict investment when creditor-shareholder conflicts become more severe.

3.2 Pooled probit

The pooled probit specification is

$$\Pr(\text{Restriction}_{it} = 1 \mid X_{it}\beta) = \Phi(X_{it}\beta), \quad (2)$$

where

$$\Phi(z) = \int_{-\infty}^z \phi(v) dv \quad (3)$$

is the standard normal cumulative distribution function.

Interpretation: The reported probit estimates are average partial effects, so they can be interpreted as changes in probability.

3.3 Random-effects probit

The random-effects probit is

$$\Pr(\text{Restriction}_{it} = 1 \mid X_{it}\beta, c_i) = \Phi(X_{it}\beta + c_i), \quad (4)$$

with

$$c_i \mid x_i \sim N(0, \sigma_c^2). \quad (5)$$

Interpretation: This specification allows persistent unobserved borrower heterogeneity, but imposes a distributional assumption on the random effect.

3.4 Fixed-effects linear probability model

The fixed-effects linear probability model can be written as

$$\text{Restriction}_{it} = \alpha_i + \lambda_t + X'_{it}\beta + u_{it}. \quad (6)$$

Interpretation: This specification absorbs time-invariant borrower heterogeneity, though it has the usual linear probability drawbacks.

3.5 Contingent-contract comparison

For borrower-lender pairs experiencing a financial covenant violation before the new agreement, the paper compares original and renegotiated contract terms:

$$\Delta Term_i = Term_i^{\text{Renegotiated}} - Term_i^{\text{Original}}. \quad (7)$$

Interpretation: Financial covenant violations are technical defaults. They give lenders acceleration rights, creating renegotiation leverage that can be used to add capital expenditure restrictions.

3.6 Investment difference-in-differences

The investment regression is approximately

$$\Delta \left(\frac{\text{CapEx}}{\text{Assets}} \right)_{it} = \alpha + \beta_N \text{NewRestriction}_{it} + \beta_O \text{OldRestriction}_{it} + \Gamma' \Delta X_{it} + \lambda_t + \epsilon_{it}. \quad (8)$$

- $\text{NewRestriction}_{it} = 1$ if the new credit agreement contains a capital expenditure restriction and the previous agreement did not.
- $\text{OldRestriction}_{it} = 1$ if the current and prior agreements both contain restrictions.
- ΔX_{it} includes changes in cash flow and market-to-book.

Interpretation: The key coefficient is β_N . A negative value means firms receiving a new restriction cut investment more than firms with no restriction.

3.7 Contractual-limit bunching test

The bunching test studies the distribution of

$$\text{Gap}_{it} = \frac{\text{CapEx}_{it} - \text{Limit}_{it}}{\text{Assets}_{i,t-1}}. \quad (9)$$

Interpretation: If capital expenditure limits bind, firms should appear disproportionately just below the contractual cap and rarely just above it.

3.8 Performance regressions

The performance regression is approximately

$$\Delta \text{Performance}_{it} = \alpha + \delta_N \text{NewRestriction}_{it} + \delta_O \text{OldRestriction}_{it} + \Pi' \Delta Z_{it} + \lambda_t + \epsilon_{it}, \quad (10)$$

where *Performance* is either market-to-book or cash flow scaled by lagged assets.

Interpretation: A positive δ_N suggests that investment restrictions may prevent inefficient investment, improving firm value and operating performance.

4 Identification and empirical design

4.1 Main identification problem

- Capital expenditure restrictions are not randomly assigned.
- Creditors impose restrictions when borrower credit quality deteriorates.
- Poor performance by itself can reduce capital expenditures, even without a contractual restriction.
- Therefore, simply comparing restricted and unrestricted firms would confound the restriction with borrower weakness.

Interpretation: The paper’s empirical challenge is to show that restrictions actually constrain investment rather than merely predict investment declines that would have occurred anyway.

4.2 Why the paper’s evidence is persuasive

The paper uses several complementary tests:

- Controls for changes in cash flow and market-to-book in investment regressions.
- Distinguishes new restrictions from old restrictions.
- Uses exact dollar limits to show bunching below the contractual cap.
- Compares the year before and the year after a newly imposed restriction.
- Examines performance after restrictions to ask whether reduced investment is value destroying or value improving.

Interpretation: No single test completely solves endogeneity, but the evidence points in the same direction: the restrictions are real constraints and are associated with improved performance.

4.3 Why renegotiation matters

- Credit agreements often contain financial covenants.
- Violating a financial covenant is a technical default.
- Technical default gives creditors bargaining power because they may accelerate the loan.
- Renegotiation after violation allows creditors to introduce new restrictions.

Interpretation: Even when contracts do not say “investment will be restricted if performance deteriorates,” the combination of financial covenants and renegotiation makes capital expenditure restrictions effectively contingent.

4.4 What the paper cannot fully prove

- It cannot randomly assign restrictions to borrowers.
- Some borrowers may reduce investment voluntarily because they are weak, even without a restriction.
- The EDGAR search does not find every private credit agreement.
- The exact dollar limit is available only for a subset of 486 contracts.
- The performance improvement could reflect other renegotiated contract changes, though the paper controls for many of them.

Interpretation: The paper's claim is strongest as a set of mutually reinforcing facts: restrictions are common, become more likely as credit quality worsens, bind actual investment, and are associated with subsequent performance gains.

5 Tables and figures

The original paper has four figures and ten table panels. The figures and tables below are directly cropped from the paper and grouped compactly to improve reading speed.

5.1 Table 1 + Table 2: sample statistics and incidence of restrictions

Read:

- Table 1 shows that 32% of credit agreements contain a capital expenditure restriction.
- In the subset with exact limits, the average restricted amount is 8.5% of lagged assets, while actual capital expenditures average 6.4% of lagged assets.
- The average agreement amount is \$450 million, and 94% of deals include a line of credit or revolver.
- Table 2 shows that 42% of firms have a restriction at some point.
- Restrictions are more common for smaller firms and worse-rated firms: only 6% for investment-grade borrowers but 43.7% for junk-rated borrowers.
- BBB borrowers have restrictions in 7.7% of agreements, BB borrowers in 39.3%, B borrowers in 49.0%, and CCC-or-worse borrowers in 62.2%.

Economic message: Creditors use direct investment restrictions frequently in private credit agreements, especially for smaller and weaker borrowers.

<i>Capital expenditure restrictions</i>				
Capital expenditure restriction (0,1)	0.319	0.000	0.466	3,720
Restriction amount/lagged assets	0.085	0.056	0.102	486
Actual capital expenditures/lagged assets	0.064	0.039	0.089	486
<i>Other loan characteristics</i>				
Agreement amount (\$ in millions)	450	200	985	3,720
Agreement amount/assets	0.338	0.245	0.308	3,720
Agreement contains a line of credit/revolver (0,1)	0.938	1.000	0.241	3,720
Agreement is secured (0,1)	0.647	1.000	0.478	3,117
Agreement contains dividend restriction (0,1)	0.813	1.000	0.390	3,446
Interest rate spread, (basis points above LIBOR)	170	150	119	3,720
Coverage ratio covenant (0,1)	0.743	1.000	0.437	3,603
Debt to cash flow covenant (0,1)	0.575	1.000	0.494	3,603
Net worth covenant (0,1)	0.452	0.000	0.498	3,603
Debt to balance sheet covenant (0,1)	0.292	0.000	0.455	3,603
Liquidity covenant (0,1)	0.147	0.000	0.354	3,603
Minimum cash flow covenant (0,1)	0.127	0.000	0.333	3,603
<i>Borrower characteristics</i>				
Cash flow/assets	0.034	0.034	0.026	3,720
Book leverage ratio	0.301	0.288	0.193	3,720
Debt to cash flow	3.052	2.260	2.888	3,526
Negative cash flow (0,1)	0.052	0.000	0.222	3,720
Financial covenant violation within past year (0,1)	0.063	0.000	0.242	3,720
Total assets (\$M)	1,622	674	1,974	3,720
Market-to-book ratio	1.768	1.426	1.136	3,720
Firm has a corporate credit rating (0,1)	0.490	0.000	0.500	3,720
<i>Conditional on borrower having credit rating</i>				
Credit rating (1 = AAA or AA, 2 = A, 3 = BBB...)	3.502	3.000	1.066	1,822
Junk rated (0,1)	0.482	0.000	0.500	1,822
CCC rated or worse (0,1)	0.020	0.000	0.141	1,822

(a) Table 1: summary statistics.

<i>Totals</i>	
Fraction of credit agreements with restriction	0.319
Fraction of firms that ever have credit agreement with restriction	0.424
Estimated fraction of all Compustat firms with restriction, using Sufi (2009)	0.336
<i>Fraction of credit agreements with restriction: By industry</i>	
Agriculture, minerals, construction	0.154
Manufacturing	0.324
Transportation, communication, utilities	0.230
Trade—wholesale	0.360
Trade—retail	0.433
Services	0.399
<i>By size (book assets)</i>	
<\$100M	0.468
\$100M–\$250M	0.469
\$250M–\$500M	0.443
\$500M–\$1,000M	0.381
\$1,000M–\$2,500M	0.243
\$2,500M–\$5,000M	0.133
>\$5,000M	0.086
<i>By debt to cash flow quartile</i>	
Debt to cash flow quartile 1	0.263
Debt to cash flow quartile 2	0.243
Debt to cash flow quartile 3	0.333
Debt to cash flow quartile 4	0.398
Cash flow negative	0.485
<i>Conditional on firm having credit rating</i>	
Investment grade	0.060
Junk rated (0,1)	0.437
AAA, AA rated (0,1)	0.000
A rated (0,1)	0.031
BBB rated (0,1)	0.077
BB rated (0,1)	0.393
B rated (0,1)	0.490
CCC rated or worse (0,1)	0.622
<i>Borrower does not have credit rating</i>	
Borrower has credit rating	0.392
	0.242

(b) Table 2: restrictions across firm types.

5.2 Table 3: credit quality and capital expenditure restrictions

Read:

- The dependent variable is whether the agreement contains a capital expenditure restriction.
- In full-sample specifications, being in the highest debt-to-cash-flow quartile raises restriction probability by about 11–22 percentage points, depending on model.
- Negative cash flow also raises the probability of a restriction.
- Financial covenant violation increases the probability in full-sample models.
- Moving from BBB to BB produces a sharp jump in restriction probability; in one specification, BB adds 23.7 percentage points relative to the omitted rating group.

Economic message: The incidence of investment restrictions is strongly tied to borrower credit risk, consistent with creditor-shareholder conflicts becoming more severe as debt becomes riskier.

	Sample (1)	Full (2)	Full (3)	Rated (4)	≥2 loans (5)	Rated and ≥2 loans (6)	≥2 loans (7)	Rated and ≥2 loans (8)
BBB rated _{t-1} (0,1)		0.026 (0.052)		0.034 (0.038)		0.030 (0.053)		0.052 (0.030)
BB rated _{t-1} (0,1)		0.237** (0.068)		0.280** (0.084)		0.215** (0.075)		0.229** (0.061)
B rated _{t-1} (0,1)		0.270** (0.083)		0.340** (0.113)		0.258** (0.095)		0.261** (0.095)
CCC rated or worse _{t-1} (0,1)		0.383** (0.143)		0.676** (0.168)		0.372** (0.172)		0.562** (0.149)
Debt to cash flow quartile 2 (0,1)	0.011 (0.026)	0.006 (0.039)	0.022 (0.030)	0.001 (0.025)	0.013 (0.031)	0.005 (0.041)	0.018 (0.027)	-0.022 (0.030)
Debt to cash flow quartile 3 (0,1)	0.088** (0.029)	0.065 (0.042)	0.126** (0.037)	0.041 (0.033)	0.085* (0.035)	0.066 (0.045)	0.055 (0.033)	-0.019 (0.039)
Debt to cash flow quartile 4 (0,1)	0.151** (0.031)	0.078 (0.044)	0.215** (0.041)	0.055 (0.034)	0.158** (0.039)	0.074 (0.047)	0.106** (0.038)	-0.007 (0.045)
Negative cash flow (0,1)	0.099* (0.045)	0.144 (0.092)	0.159** (0.061)	0.131 (0.103)	0.158** (0.066)	0.167 (0.118)	0.170* (0.074)	0.223 (0.125)
Financial covenant violation _{t-1} (0,1)	0.126** (0.040)	0.069 (0.060)	0.171** (0.052)	0.056 (0.047)	0.130** (0.050)	0.074 (0.074)	0.147** (0.047)	0.133* (0.067)
Ln(total assets _{t-1} (\$M))	-0.071** (0.007)	-0.046** (0.012)	-0.085** (0.008)	-0.032** (0.009)	-0.083** (0.009)	-0.042** (0.013)	-0.053 (0.027)	0.025 (0.043)
Market-to-book ratio _{t-1}	-0.024** (0.009)	-0.030 (0.016)	-0.029** (0.010)	-0.026* (0.010)	-0.028* (0.012)	-0.028 (0.020)	-0.018 (0.012)	-0.042** (0.020)
Number of credit agreements	3,720	1,822	3,720	1,822	2,742	1,453	2,742	1,453
Number of firms	1,939	844	1,939	844	961	475	961	475
R ²	0.17	0.30	0.13	0.26	0.20	0.30	0.19	0.21

*,** Statistically distinct from zero at the 5% and 1% level, respectively.

Table 1: Table 3: credit quality and capital expenditure restrictions.

5.3 Table 4 + Table 5: restrictions versus other loan terms

Read:

- Table 4 shows that capital expenditure restrictions are positively correlated with secured status, interest spreads, dividend restrictions, and several financial covenants.
- The restriction is especially correlated with secured loans and interest spreads.
- Table 5 shows that the capital expenditure restriction becomes much more likely as ratings deteriorate.
- From A-or-better to BBB, the mean restriction rate rises from 2.8% to 7.7%; at BB it jumps to 39.3%.
- The sensitivity of investment restrictions to rating deterioration is comparable to or larger than the sensitivity of many standard loan terms.

Economic message: Capital expenditure restrictions are not peripheral boilerplate. They respond to borrower risk in the same way as spreads, collateral, and other core contractual protections.

	restriction	spread	covenant	covenant	covenant	balance sheet	covenant	covenant
Capital expenditure restriction	0.26**	0.42**	0.41**	0.15**	0.19**	0.08**	-0.18**	0.00
Dividend restriction								0.25**
Secured	0.38**	0.33**	0.19**	0.20**	0.13**	-0.14**	0.10**	0.12**
Interest spread		0.59**	0.14**	0.14**	0.09**	-0.24**	0.17**	0.21**
Coverage covenant			0.06**	0.07**	0.00	-0.22**	0.14**	0.29**
Debt to cash flow covenant				0.38**	0.03	-0.10**	-0.07**	-0.08**
Debt to balance sheet covenant					0.05*	-0.41**	-0.10**	-0.01
Net worth covenant						-0.03	0.10**	-0.02
Liquidity covenant							0.03	-0.13**
Cash flow only covenant								0.09**

(a) Table 4: correlations across contract terms.

Panel A: Means by credit rating					
	A or better	BBB	BB	B	CCC or worse
N	322	622	527	314	37
Capital expenditure restriction	0.028	0.077*	0.393*	0.490*	0.622*
Interest spread	48	98*	195*	258*	362*
Dividend restriction	0.390	0.602*	0.940*	0.936*	0.875*
Secured	0.065	0.218*	0.808*	0.961*	1.000*
Coverage covenant	0.387	0.695*	0.859*	0.764*	0.528
Debt to cash flow covenant	0.201	0.440*	0.686*	0.623*	0.361
Net worth covenant	0.217	0.325*	0.397*	0.306	0.167
Debt to balance sheet covenant	0.481	0.454	0.211*	0.094*	0.056*
Liquidity covenant	0.022	0.018	0.138*	0.148*	0.222*
Cash flow only covenant	0.013	0.023	0.107*	0.209*	0.472*

Panel B: Percentage change relative to the mean from A or better				
	BBB	BB	B	CCC or worse
Capital expenditure restriction	15%	115%	145%	186%
Interest spread	29%	86%	123%	183%
Dividend restriction	26%	68%	67%	60%
Secured	24%	115%	139%	145%
Coverage covenant	42%	64%	51%	19%
Debt to cash flow covenant	42%	84%	73%	28%
Net worth covenant	24%	40%	20%	-11%
Debt to balance sheet covenant	-9%	-93%	-132%	-146%
Liquidity covenant	-3%	79%	86%	136%
Cash flow only covenant	8%	74%	155%	362%

* Statistically distinct from an A-rated firm or better at a 5% level.

(b) Table 5: contract terms by credit rating.

5.4 Table 6: renegotiation after financial covenant violations

Read:

- Among contingent contract pairs, the fraction with capital expenditure restrictions rises from 40.0% in the original agreement to 60.4% in the renegotiated agreement.
- This is a 51% increase.
- Interest spreads rise from 196 to 255 basis points, a 30% increase.
- Secured status rises from 79.5% to 95.7%, a 20% increase.
- The elasticity of the capital expenditure restriction is larger than for many other contract terms.

Economic message: Financial covenant violations give creditors control rights. Renegotiation turns standard financial covenants into a mechanism for imposing new investment restrictions after bad performance.

	(1) Capital expenditure restriction	(2) Interest spread	(3) Secured	(4) Dividend restriction
Original agreement	0.400	196	0.795	0.864
Renegotiated agreement that follows financial covenant violation	0.604*	255**	0.957**	0.940
% Increase	51%	30%	20%	9%
Number of credit agreements	103	103	90	95
Number of firms	48	48	46	48

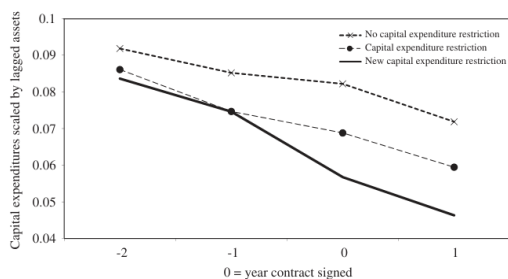
Table 2: Table 6: contingent contract analysis.

5.5 Figure 1 + Table 7: the effect of restrictions on investment

Read:

- Figure 1 shows that capital expenditures decline for all firms around loan origination, but the decline is larger for firms receiving a new capital expenditure restriction.
- New-restriction firms fall from roughly 8.3% of assets two years before the contract to roughly 4.6% one year after.
- Table 7 confirms this in regressions: new restrictions reduce changes in capital expenditures by about 0.9–1.4 percentage points of lagged assets.
- The effect remains after controlling for changes in cash flow and market-to-book.

Economic message: The restrictions are not merely descriptive. They are associated with a statistically meaningful reduction in actual investment.



itures, by whether contract contains capital expenditure restriction. This figure presents average capital expenditures, by whether the credit agreement contains a covenant with a direct restriction on capital expenditures.

(a) Figure 1: capital expenditures by restriction status.

	year before contract–year of contract		average two years before contract–average year of contract	
Old capital expenditure restriction	–0.003 (0.002)	–0.003 (0.002)	–0.004 (0.003)	–0.004 (0.002)
New capital expenditure restriction	–0.014** (0.005)	–0.009* (0.004)	–0.014** (0.005)	–0.010* (0.005)
Difference cash flow to lagged assets		0.126** (0.016)		0.211** (0.021)
Difference market-to-book		0.016** (0.002)		0.005* (0.003)
N	2,812	2,812	2,812	2,812
R ²	0.01	0.16	0.03	0.14

*,** Statistically distinct from zero at the 5% and 1% level, respectively.

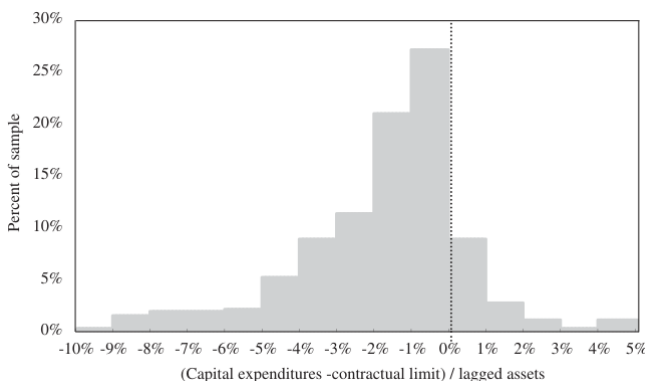
(b) Table 7: investment regressions.

5.6 Figure 2 + Table 8: clustering below the contractual limit

Read:

- Figure 2 plots actual capital expenditures minus the contractual limit, scaled by lagged assets.
- The distribution bunches just below zero.
- The paper reports that 27% of observations are within 1% of assets below the limit, while only 9% are within 1% above it.
- Table 8 shows that the bunching pattern is similar whether the contract is signed six or more months before year-end or less than six months before year-end.

Economic message: The bunching evidence is strong evidence that the contractual cap is binding. If the cap were merely recording planned spending, the distribution should not show such asymmetric mass just below the limit.



(a) Figure 2: expenditures minus contractual limit.

lagged assets	signed > 6 months prior to first year that limit applies	signed < 6 months prior to first year that limit applies
Below –10%	0.030	0.050
(–10%, –9%]	0.000	0.000
(–9%, –8%]	0.010	0.020
(–8%, –7%]	0.030	0.010
(–7%, –6%]	0.020	0.020
(–6%, –5%]	0.020	0.020
(–5%, –4%]	0.060	0.050
(–4%, –3%]	0.070	0.110
(–3%, –2%]	0.120	0.100
(–2%, –1%]	0.210	0.210
(–1%, 0%]	0.260	0.290
(0%, 1%]	0.090	0.080
(1%, 2%]	0.030	0.030
(2%, 3%]	0.010	0.010
(3%, 4%]	0.000	0.000
(4%, 5%]	0.010	0.010
Above 5%	0.020	0.010

*,** Statistically distinct from column (1) at the 5% and 1% percent level, respectively.

(b) Table 8: timing from contract to year-end.

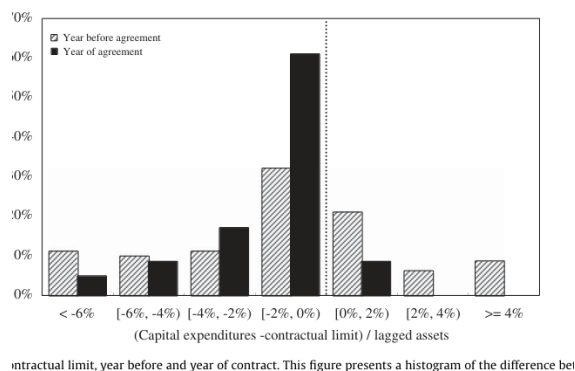
5.7 Figure 3 + Figure 4: before-after bindingness and performance

Read:

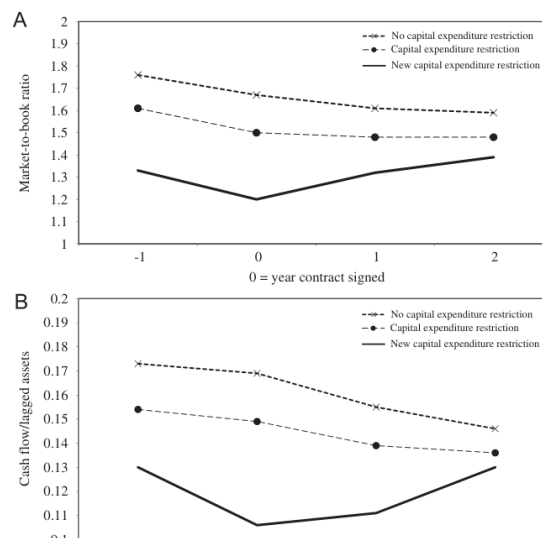
- Figure 3 focuses on 81 loans where the current agreement imposes a restriction but the prior agreement did not.
- Before the new restriction, many firms exceed the future limit. After the restriction, firms concentrate just below the limit.

- The paper reports that over 60% of observations lie within 2% of assets just below the threshold after the restriction is imposed.
- Figure 4 shows market-to-book and cash-flow performance before and after loan origination.
- Firms with a new restriction have lower pre-contract performance, but market-to-book and cash flow improve after the new restriction.

Economic message: The before-after histogram directly supports the idea that the new limit changes behavior. The performance figure suggests the investment reduction may prevent inefficient spending rather than simply choke off good projects.



(a) Figure 3: year before versus year of contract.



(b) Figure 4: market value and operating performance.

5.8 Table 9 + Table A1: performance regressions and sample selection

Read:

- Table 9 shows that a new capital expenditure restriction is associated with an increase in market-to-book of about 0.17.
- It is also associated with an increase in cash flow scaled by lagged assets of about 0.012–0.016.
- These estimates remain after controlling for changes in cash flow, market-to-book, and other loan terms.
- Table A1 addresses EDGAR sample-selection concerns.
- In a random sample of 200 observations not found by the search program, 81 full contracts are in EDGAR but missed, 45 are minor amendments, and 74 are not in EDGAR.

Economic message: The performance regressions strengthen the interpretation that creditor restrictions can create positive externalities for equityholders by constraining inefficient investment.

Old capital expenditure restriction	0.023 (0.030)	0.098 (0.030)	0.021 (0.033)	0.006 (0.003)	0.009 (0.004)	0.002 (0.005)
New capital expenditure restriction	0.171** (0.042)	0.167** (0.039)	0.170** (0.049)	0.016** (0.006)	0.012* (0.006)	0.014* (0.006)
Difference cash flow to lagged assets		-0.138 (0.181)	-0.138 (0.184)		-0.249** (0.033)	-0.249** (0.033)
Difference market-to-book ratio		-0.185** (0.036)	-0.187** (0.036)		0.032** (0.004)	0.032** (0.004)
Old dividend restriction			0.013 (0.031)		0.002 (0.004)	
New dividend restriction			-0.011 (0.045)		0.004 (0.005)	
Old secured			0.023 (0.032)		0.000 (0.004)	
New secured			0.008 (0.058)		-0.011 (0.006)	
Old coverage covenant			-0.029 (0.031)		-0.007 (0.004)	
New coverage covenant			0.004 (0.056)		0.001 (0.008)	
Old debt to cash flow covenant			-0.085** (0.027)		-0.014** (0.004)	
New debt to cash flow covenant			-0.019 (0.049)		-0.019** (0.007)	
Old net worth covenant			0.013 (0.027)		-0.002 (0.004)	
New net worth covenant			-0.124 (0.086)		-0.011 (0.009)	
Old liquidity covenant			-0.009 (0.051)		0.003 (0.007)	
New liquidity covenant			0.144 (0.105)		0.004 (0.015)	
Old cash flow covenant			0.013 (0.055)		0.011* (0.007)	
New cash flow covenant			0.008 (0.069)		0.020** (0.008)	

(a) Table 9: value and operating performance.

Random sample of 200 observations not in sample					
Full contract in EDGAR but search program misses it					
Minor amendment in EDGAR					
Not in EDGAR					
Panel B: Comparison of characteristics					
	(1)	(2)	(3)	(4)	(5)
	In sample (N = 3,720)	Not in sample (N = 200)	Not in sample, full contract in EDGAR (N = 81)	Not in sample, minor amendment in EDGAR (N = 45)	Not in sample, contract not in EDGAR (N = 74)
Capital expenditure restriction	0.329		0.370		
DealScan data quality					
Secured not available	0.162	0.394**	0.147	0.409**	0.652**
Financial covenant not available	0.069	0.383**	0.147	0.341**	0.667**
% Held by lenders not available	0.568	0.702**	0.613	0.682**	0.812**
Div restriction not available	0.074	0.473**	0.293**	0.386**	0.725**
Type of loan					
Sole-lender loan	0.182	0.394**	0.667**	0.227	0.203
Revolver w/maturity of <one year	0.096	0.250**	0.093	0.227*	0.435**
Borrower characteristics					
Total assets (\$M)	1623	1589	631**	1544	2658**
Book debt/total assets	0.300	0.291	0.282	0.327	0.278
Cash flow/total assets	0.034	0.026**	0.019**	0.030	0.032
Loan characteristics					
Loan amount (\$M)	450	340*	145**	358	541
Interest rate spread (basis points)	170	198*	261**	190	134*

*,** Statistically distinct from In sample category at 5% and 1% level, respectively.

(b) Table A1: sample-selection checks.

6 Short summary

- The paper studies whether private creditors directly restrict firm investment.
- The sample contains 3,720 private credit agreements for 1,931 public U.S. firms from 1996 to 2005.
- 32% of the agreements contain explicit capital expenditure restrictions.
- Restrictions are more common among smaller firms, firms with worse credit quality, and firms with poor cash-flow measures.
- A downgrade from BBB to BB produces a large jump in the probability of a restriction.
- Capital expenditure restrictions respond to credit quality at least as strongly as other standard loan terms.
- Financial covenant violations help make restrictions state-contingent because technical default gives creditors renegotiation rights.
- In contingent contract pairs, restrictions rise from 40.0% to 60.4% after covenant violations.
- Firms receiving new restrictions cut capital expenditures more than firms without restrictions.
- Exact contractual limits reveal bunching just below the spending cap, showing that restrictions bind.
- Before a new restriction, many firms spend above the future cap; after the restriction, most observations cluster below the cap.
- Firms receiving new restrictions subsequently show increases in market-to-book and operating performance.
- The paper concludes that creditor control rights can shape real firm investment and may improve efficiency by limiting overinvestment or risk-shifting.

7 Conclusion

7.1 Core conclusion

Nini, Smith, and Sufi show that direct restrictions on firm investment are common in private credit agreements and that these restrictions matter for real corporate policy. The central empirical result is that creditor control rights become more likely as borrower credit quality deteriorates, constrain capital expenditures when imposed, and are associated with subsequent improvements in performance.

7.2 Economic intuition

When borrowers become riskier, creditors worry that equityholders may undertake investments that shift risk or destroy total firm value while benefiting equity. Private credit agreements give creditors control rights through covenants and renegotiation. Capital expenditure restrictions are one direct way to limit those investment choices.

7.3 Takeaway

The paper changes the interpretation of private debt contracts. Bank loans are not just financing instruments with interest rates and collateral. They are governance contracts that allocate control rights over real decisions. Investment policy can therefore be shaped not only by internal cash flow and investment opportunities, but also by creditor-imposed contractual restrictions.